

# Text-Based Speaker Identification for Video Game Dialogues

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**Abstract.** Speaker identification presents a challenging task in the field of natural language processing and is believed to be an important step in building believable human-like systems. Although most of the existing work focuses on utilizing acoustic features for the task, in this paper we propose a text-dependent transformer-based machine learning approach. Our research shows promising results improving on the existing work that utilized textual features for the task, while simultaneously introducing a new dataset acquired from a commercial video game. To our knowledge, this is the first time text-dependent speaker identification was performed in the domain of video games.

**Keywords:** Speaker identification · Video games · Dialogue · BERT

## 1 Introduction

Speaker identification is the task of connecting an utterance to the speaker that produced it. It can be used in the development of dialogue systems with some authors citing it as a pivotal step in the development of believable human agents ([6], 2003). Given a set of possible speakers, the problem of speaker identification can be formulated as a simple classification problem: which utterance belongs to a certain class, or rather, to a certain speaker.

Humans tend to perform speaker identification unconsciously, gathering audiovisual information, such as the direction from which the sound came or even vocal timbre, in order to connect utterances to a speaker. In modern days, when using textual communication through chat applications or text messages, information like emojis and stickers can sometimes help us identify the sender just from the content of the message. Additionally, characteristics such as style of speech or common phrases are also used to determine the speaker.

It should be noted that when analyzing daily conversations, we do require a large amount of information to identify a certain speaker from a typical phrase, e.g. to establish their style of speech. However, storytelling mediums, like movies, TV shows, books or video games, especially those set in the realm of fantasy,

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often accentuate these idiosyncrasies in order to increase the immersion experience and make the story more believable.

Here we present our work that consists of performing two different subtasks of speaker identification: the first being the classification of utterances to a single speaker, focusing on character-specific traits, and the second being the mapping of utterances to a class of speakers, in turn detecting class-specific traits. For this we used two datasets, one that is publicly available as part of the LIGHT research platform ([16], 2019), and another one gathered from a commercial video game *Dragon Age: Origins* using publicly available tools to acquire the said data. Due to the large amount of utterances that are available per speaker, we decided to use the dataset acquired from *Dragon Age: Origins* to identify a single speaker, and LIGHT dataset to identify the class of the speaker. Our approach showed an accuracy of 90.27% when identifying class of speaker and of 74.86% on a character identification level, outperforming methods used as a baseline.

The contributions of our work are listed as follows:

- We proposed a simple yet effective transformer-based approach for speaker identification that utilizes textual data only.
- To our knowledge, our work is the first of its kind to examine the possibility of using video games as a source of data for the speaker identification task, further opening the possibility of using interactive storytelling mediums for the said task.
- Although there has been some previous work in creating a video game related corpora for NLP tasks from commercial video games ([1], 2020), we created the first dataset centered around dialogues in this domain which can be found at the following GitHub link: https://github.com/dradisavljevic/DAODataset.
- The proposed method is transferable to transcripts of real life conversations.

The paper is structured as follows. Section 2 presents previous work done in the area of text-based speaker identification. In Sect. 3 we give an overview of the datasets used for our evaluation. Section 4 presents the methods used in this paper while Sect. 5 describes the experiments and lists the results. Section 6 discusses both our findings and results listed in Sect. 5. Finally, Sect. 7 concludes this paper and provides our plans for future work.

## 2 Related Works

Most of the work in the area of speaker identification is centered around speech or signal processing ([14], 2019), however, the work of [10] (2007) has shown that textual features can also reflect the personality of the speaker, suggesting that a text-dependent approach can be beneficial to speaker identification. To the best of our knowledge, experiments that utilize textual features only for the task of identification are few in number ([12], 2015; [8], 2012; [9], 2017).

[12] proposed using logistic regression and recurrent neural networks for the task of detecting change in speaker through turn taking in movie dialogues.

The task itself is different from ours as it focuses on detecting turn changes rather than identifying actual speakers. Another work that focuses on movie dialogues is that of [8], which was used as a starting point for our experiments. They have suggested using various classification algorithms, such as Naive Bayes (NB), K-Nearest Neighbors (KNN) and Conditional Random Forest (CRF) on vectors formed out of certain stylistic features, such as the number of adverbs or adjectives per words in each utterance, in order to identify the speaker on a dataset acquired from the movie script database[1]. The work of [9] used these results as a baseline and conducted additional experiments with Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN) on transcripts from the TV show *Friends*, reporting improved results with CNN. However, to the best of our knowledge, no research has been conducted on text-dependent speaker identification in the domain of video games.

## 3 Datasets

For the task of speaker identification, two video game-related datasets are used:

1. Dataset composed of textual dialogues extracted from the *Dragon Age: Origins*[2] video game developed by *Bioware* studios.
2. Dialogue data extracted from a large scale, crowdsourced, text adventure game dataset, developed as part of the LIGHT research platform which comes from the work of [16].

### 3.1 Dragon Age: Origins Dialogue Dataset

*Dragon Age: Origins* is an action role-playing video game developed by *Bioware* studios in 2009. It features a large amount of dialogue between player and NPCs[3] that have different personalities, thus making it suitable for the task of speaker identification. Text data was obtained using two resources: *Dragon Age Wiki*[4] a fan website that contains information related to the video game series and *Dragon Age Toolset*[5], software released by developers of the game in order to allow modification of in-game content.

*Dragon Age Toolset* comes with the Microsoft SQL Server relational database containing all the video game resources, including dialogue utterances in textual form. Utterances were extracted from the database, preprocessed and stored into a dialogue file. Each utterance has a speaker and a main listener (the persona to whom the utterance is spoken), as well as an identifier of the setting under which the utterance occurs (part of a quest, a random encounter, etc.). All the missing information was filled in manually by referencing the *Dragon Age Wiki* website.

[1] http://www.imsdb.com/.
[2] https://www.ea.com/games/dragon-age/dragon-age-origins.
[3] Non Playable Character.
[4] https://dragonage.fandom.com/.
[5] https://dragonage.fandom.com/wiki/Toolset.

We separated and labeled utterances that are spoken by the player and ten of their NPC companions. Utterances that belong to remaining characters in the game are all labeled as 'Others', as they are too few in number per character. Figure 1 gives a graphical representation of the dialogue distribution within the dataset.

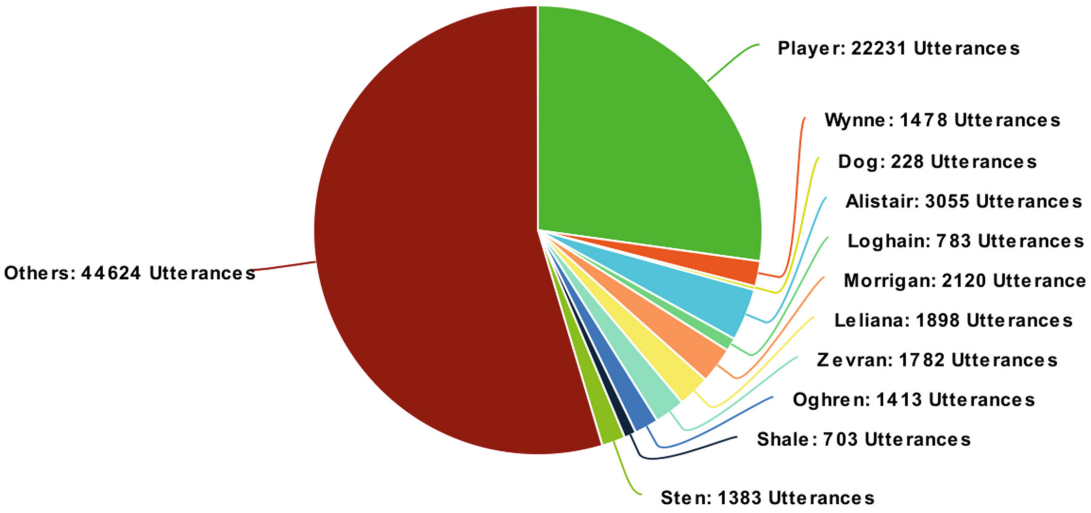



**Fig. 1.** Number of utterances obtained from the *dragon age: origins* per character

### 3.2 LIGHT Research Platform Dataset

LIGHT is a fantasy text adventure game research platform designed for studying grounded dialogue. The dataset contains a large set of crowdsourced interactions (about 11,000 episodes) which are made up of actions and dialogue utterances. Data is publicly available through the ParlAI[6] platform. For this work we only considered the episodes that are part of the original LIGHT dataset. All the utterances from the episodes are extracted, preprocessed and stored into a dialogue file.

LIGHT dataset contains a large amount of characters (940 characters with utterances), and unlike the other dataset we utilized, lacks a clear main character as it does not follow a linear story. Additionally, most of the characters present in the dataset have a character class assigned to them, with about 20% of them belonging to an undefined class. For unlabeled characters, we have manually assigned them one of the three existing classes: object, creature or person, based on their persona description which is also given in the dataset.

Due to the above mentioned reasons we turned to predicting character class from utterance rather than predicting a single character for this dataset. Figure 2 shows utterance distribution per character class in the LIGHT dataset.

[6] http://parl.ai/projects/light.

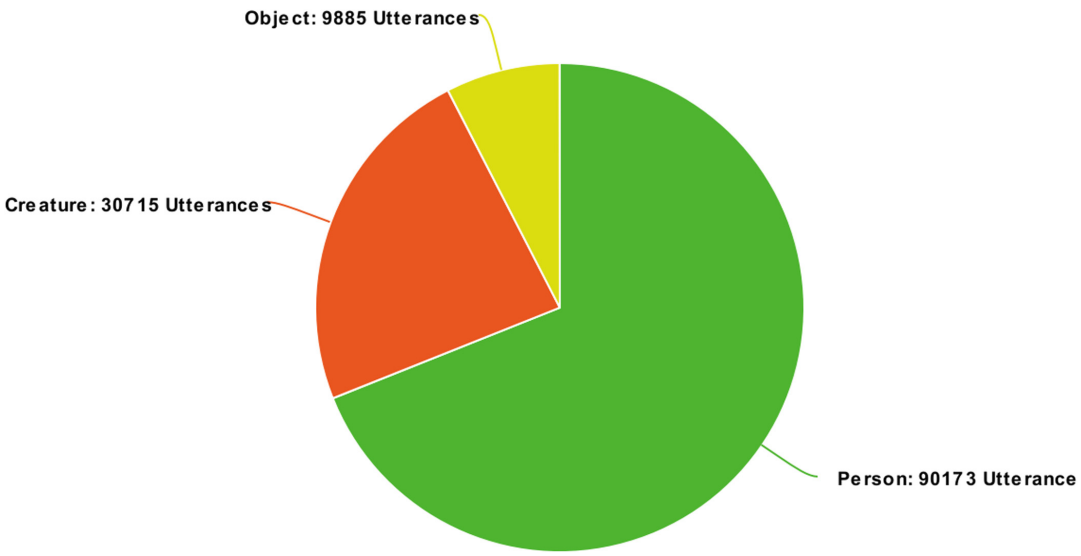



**Fig. 2.** Number of utterances obtained from the LIGHT project dataset per character class

## 4 Methods

We have evaluated four algorithms for the task of speaker identification. Methods proposed in the work of [9] and [8] were used as baseline for our approach that is based on Bidirectional Encoder Representations from Transformers (BERT) ([4], 2019).

### 4.1 K-Nearest Neighbors

In the work of [9] KNN was used as a baseline approach for speaker identification, while in the work of [8] it was the best performing algorithm for the same task. Although other approaches proposed by [9] outperform the KNN method, we have still decided to use it as a baseline due to its stability and simple implementation.

The algorithm has been implemented using the Faiss[7] library in Python programming language for speed and efficiency. The implementation followed the specifications given by [8], with same stylistic features being used in combination with cosine similarity as a distance function.

### 4.2 Convolutional Neural Network

[9] have reported satisfactory results using CNN model proposed by [7] (2014) with minor modifications. This approach is selected as another baseline for our experiments. The model consists of a single convolution layer, followed by a global max-pooling layer and a fully-connected layer with a softmax function

[7] https://github.com/facebookresearch/faiss.

used to normalize the output. For the sake of simplicity, we have decided not to enforce the Euclidean ($L_2$) norm constraint as the work of [17] (2016) found that it has little effect on the final result.

Unlike the previous work, we have decided to add an embedding layer instead of using the pretrained word embeddings, due to a large amount of colloquialisms and fantasy-related vocabulary present in the data, as well as the size of the data itself. This has made the training phase slower compared to previous work, as we train the embeddings as well. In order to avoid possible overfitting, we used early stopping ([11], 1998).

### 4.3 Convolutional Neural Network with Utterance Concatenation

The final baseline method is also based on the proposal of [9] and is an approach that involves using CNN while grouping the utterances of a single character within a single scene, in chronological order. [9] have reported that this approach achieves improved F1 and accuracy score on the speaker identification task. We have replicated the process by using a single episode from the LIGHT dataset as an equivalent to one TV series scene. For the dataset that comes from the *Dragon Age: Origins* video game, we considered utterances that have the same 'speaker' and 'settingname' attribute. Utterances that share the same value for the 'settingname' attribute belong to the same dialogue tree[8], which makes them an equivalent to a TV series scene.

[9] have also improved the method by restricting the prediction search space to the set of speakers present in the scene. However, due to the majority of video game dialogues being limited to only 2 participants, we have decided not to follow this approach.

### 4.4 BERT

BERT is a transformer-based model that was published in 2018 and has been used ever since for a variety of NLP tasks, achieving considerably better results than other approaches, especially for classification problems ([13], 2019; [5], 2020). The work of [2] (2019) reported satisfactory results when it comes to multi-label text classification on documents, which inspired us to test it on a single utterance level.

For both the speaker and speaker class identification, we used uncased BERT base model that utilizes Adam optimization algorithm with weight decay as an optimizer. The experiments were run on a single utterance level and with concatenated texts for both of the datasets through 5 epochs. The dropout was set to 0.1 and learning rate to 1e-5. Due to memory constraints, we used a batch size of 3.

## 5 Experiments

For our experiments, all the utterances extracted from both datasets are first cleaned and tokenized. With KNN-based and CNN-based methods we have used a

[8] https://en.wikipedia.org/wiki/Dialogue_tree.

custom tokenizer that is a minor modification of that used in the work of [7], while for the BERT model we have used a BERT tokenizer. For the KNN method, 70% of the data was taken as a training set and 30% as an evaluation set. For other approaches, 70% of the data was used for training, while 15% was used for validation and the remaining 15% for evaluation set. Due to the nature of the data split, experiments were conducted over 10 trials, taking the mean value as the result.

We have experimentally established that for KNN approach, results are best for k=13 on *Dragon Age: Origins* dataset, while for LIGHT dataset the results are best for k=15.

Figures 1 and 2 show that there is a severe class imbalance present in both of the datasets. This led us to taking an oversampling approach ([3], 2002) for KNN and CNN based methods in order to prevent classes with a large amount of examples skewing the classifier output. However, since BERT has been shown to perform well even on an imbalanced dataset ([15], 2019), in this work we report the results it achieved without the use of any data augmentation techniques.

For the training phase of the CNN we used a batch size of 100 and used early stopping to determine the correct number of epochs needed for the training process without overfitting. For the BERT approach, we used 5 epochs. Table 1 displays our results with the upper half showing results on the *Dragon Age: Origins* dataset for speaker identification task and the lower half reporting results on the LIGHT dataset for the speaker class identification task.

## 6 Discussion

In the work of [8] KNN was the best performing algorithm on a movie script dataset, with an accuracy of 30.39%, while other metrics were not reported. The experiments of [9] had the most success using CNN in combination with

**Table 1.** Performance per model. **MF1** stands for macro F1 score and **WF1** for F1 score weighted by number of true labels per class.

| Model | Precision | Recall | Accuracy | MF1 | WF1 |
|---|---|---|---|---|---|
| Dragon age: Origins dataset | | | | | |
| KNN | 3.60 | 10.33 | 20.05 | 4.61 | 8.07 |
| CNN | 43.66 | 36.25 | 39.66 | 36.79 | 37.30 |
| CNN-Concatenation | 36.33 | 27.96 | 45.67 | 24.85 | 34.78 |
| BERT | 46.43 | 52.76 | **74.86** | 49.06 | **74.23** |
| BERT-Concatenation | **60.58** | **64.42** | 70.21 | **50.55** | 69.3 |
| LIGHT dataset | | | | | |
| KNN | 38.58 | 33.74 | 76.33 | 30.12 | 67.00 |
| CNN | **76.02** | 52.68 | 79.91 | 49.85 | 72.48 |
| CNN-Concatenation | 53.17 | 49.34 | 83.29 | 50.68 | 62.12 |
| BERT | 48.83 | 63.67 | 83.42 | 50.88 | 82.59 |
| BERT-Concatenation | 70.67 | **73.02** | **90.27** | **71.71** | **90.19** |

utterance concatenation while also restricting the set of prediction labels to speakers present in the scene on the dialogues acquired from the transcripts of the TV show *Friends*. They reported an accuracy of 46.48% and an F1 score of 44.19%.

Using KNN and CNN on the *Dragon Age: Origins* dataset proved to be less successful, with CNN yielding the best results out of all the baseline methods. It should however be noted that, surprisingly, concatenated utterances led to an increase in the number of false positives and false negatives (lower recall and precision values) rather than improving the basic CNN approach. After looking at the results, we believe that this is due to concatenated utterances being more difficult to discern one from another, as they share from a common pool of words and topics.

When it comes to the experiments we conducted on the LIGHT dataset, the results achieved are better than those using the same methods on the *Dragon Age: Origins* dataset. This is to be expected due to the smaller number of labels present for this subtask, making it easier to connect an utterance to character class than to a single character. We would like to note that, even though using CNN with concatenated utterances led to higher accuracy, it also led to an increase in both false positives and false negatives (reduction in precision and recall). Although this was surprising, since we tried to balance out the classes for the experiment, it is our understanding that synthetic examples led to generalization which in turn resulted in a slightly worse than expected performance.

According to Table 1, BERT outperforms baseline methods without any data augmentation approaches. However, BERT still produces a considerable amount of false positive and false negative predictions per class, as indicated by the low precision and recall. We believe that this can be improved by using data augmentation techniques in the future alongside the BERT model. Figure 3 is a confusion matrix for characters present in the *Dragon Age: Origins* dataset. We can see from the confusion matrix that despite them originating from the class with fewest utterances, predictions for the 'Dog' character label achieve an accuracy of almost 100%, due to these utterances being specific. We believe that this demonstrates good performance of the model on sets of utterances that are less generic in nature.

It is interesting to note that characters that this model tends to confuse the most are those that express some shared personality traits. For example, both the 'Leliana' and 'Alistair' are stereotypical good characters that disapprove of 'Player' character acting in an evil way, and the model has proportionally shown the second highest confusion in prediction between the two of them.

The highest confusion for the model is between the 'Player' and 'Others' character label. We believe that this is due to a wide variety of utterances that are available as dialogue options for the 'Player' character. Some of them are mutually exclusive, meaning that if a player was to select one utterance as a dialogue option, other utterances that appeared as options would not come up in the game. This gives the players possibility of reflecting different personality traits through dialogue. However, since these utterances were not separated in

the dataset for the experiments performed in this work, the 'Player' character most likely expressed multiple personality traits, some even conflicting. This most likely led the model to confusing the 'Player' and 'Others' category, as the characters in the 'Others' category come from different backgrounds and themselves exhibit a wide variety of personality traits.

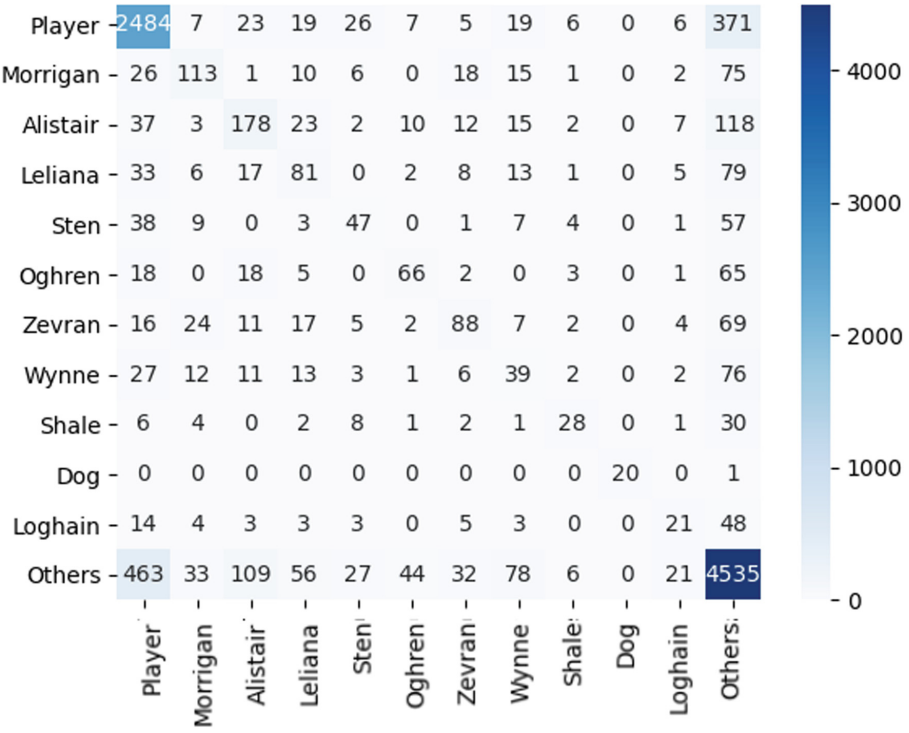



**Fig. 3.** Confusion matrix for *dragon age: origins* dataset on a single utterance level

## 7 Conclusion and Future Work

In this paper we have proposed a new, transformer-based, machine learning approach for the speaker identification task. Even though most of the works for the task deal with audio and signal processing, for our work we opted to use textual data only. In order to determine how well transformer-based approach compares to methods described in other studies that have utilized textual data, we have replicated experiments described in them and used them as a baseline for our method. Our experiments have confirmed the findings from previous works by other authors [9] that contextual information is important for speaker identification, even in the domain of video game dialogues. Comparing the performance of different approaches we have seen an increase in most of the metrics when using utterance concatenation in order to provide more context.

The corpora proved to be a good starting point for the task. However, additional steps can be taken in the future to further improve the results. For example an increase in size should lead to improved results, but combining different

commercial video games into a single corpora might lead to some level of confusion for the model. This is due to certain character tropes being present across multiple different works of fiction (stereotypical villain phrases for example). Additionally, character classes present in the LIGHT dialogue dataset could be more finely granulated. This might create difficulties for the model to predict the correct class, as they would increase in number, but it can be interesting to ascertain any similarities between characters from the same background (e.g. elven race, warrior background, etc.).

Results from our experiments suggest that it is possible to determine which utterance belongs to a certain character with high probability in the domain of video game dialogues. Due to the entertaining nature of the domain, certain traits tend to be overly exaggerated which might assist the transformer-based model to predict correctly. Observing the BERT based method on transcripts of real life conversations could also prove to be interesting. Additionally, because the dialogue dataset used for our experiments comes from a commercial video game that does not rely on text as a sole medium of storytelling, we believe that this approach can also yield good results on games that are more text-driven in nature.

Although the combination of audio and visual data might be a more natural source for speaker identification, we have shown that textual data can be used with some success. It is our hope that this will lead to increased usage of textual data for the same task, and inspire more research on datasets that come from interactive storytelling mediums.

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